Lab 9: Software Architecture Design & Design Patterns

Aim

To design the software architecture of the AI-Based Interview Preparation Platform and identify suitable design patterns.

1. System Architecture

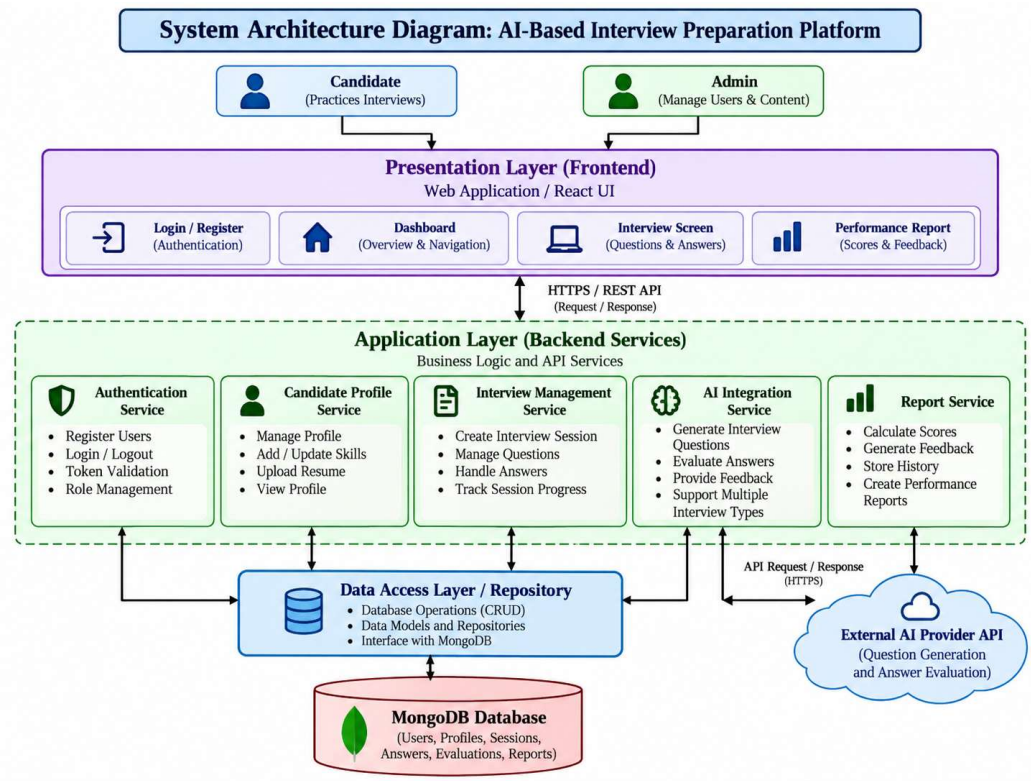


Figure 1: System Architecture of AI-Based Interview Preparation Platform.

2. Architecture Components

Component	Responsibility
Presentation Layer	Displays login, dashboard and interview screens.
Authentication Service	Handles registration and login.

Interview Management	Manages interview sessions, questions and answers.
AI Integration	Generates questions and evaluates answers using an AI service.
Data Access Layer	Handles database operations.
MongoDB	Stores users, sessions, answers and results.
Report Service	Generates feedback and performance reports.

3. Data Flow

1. Candidate logs in to the platform.
2. Candidate starts a mock interview.
3. The AI service generates questions.
4. Candidate submits answers for evaluation.
5. Results are stored in MongoDB.
6. The platform displays feedback and a performance report.

4. Design Patterns

Pattern	Application
MVC	Separates user interface and application logic.
Strategy	Supports different evaluation strategies for HR, technical and coding interviews.
Factory	Creates interview objects based on interview type.
Adapter	Integrates external AI providers through a common interface.
Repository	Separates database operations from business logic.

5. Advantages

- Modular and maintainable architecture.
- Easier testing and debugging.
- Flexible AI service integration.
- Organized database operations.
- Easier addition of new interview types.

Conclusion

The proposed architecture separates the presentation layer, backend services, AI integration and database. Suitable design patterns improve modularity, flexibility and maintainability.

Result: The software architecture and suitable design patterns for the AI-Based Interview Preparation Platform were studied.